FIITJEE

ALL INDIA INTEGRATED TEST SERIES

PART TEST – III

JEE (Advanced)-2021

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

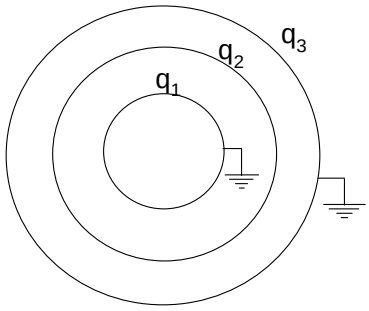
Physics

PART – I

SECTION – A

1. ABC

Sol.



2. ABCD

Sol. $q_P = \frac{qE}{m} = \frac{q\alpha}{m} - \left(\frac{q\beta}{m}\right)x$

$q_{P_0} = \left(-\frac{q\beta}{m}\right)x$

Hence,
Motion is oscillatory.
For mean position,
$F = 0$

$x = \frac{\alpha}{\beta}.$

Also, $A = \frac{\alpha}{\beta}.$

and Max. acceleration $= \left(\frac{q\beta}{m}\right)(A) = \frac{q\alpha}{m}.$

3. ABD
Sol. Use concept of induced electric field.

4. ABC
Sol. Use $\vec{F} = q\left(\vec{v} \times \vec{B}\right)$

5. ABCD
Sol. At t = 0, capacitor behaves ac short circuited and at steady state, it behaves as open circuited.

6. AC
Sol. Use B due to straight current carrying wire and due to a circular wire.
Also use torque on closed 100 p carrying current placed in a magnetic field.

7. AD
Sol. Concept of emf and potential difference.

8. B
Sol. Motion of charged particle in a magnetic field $\vec{F} = q\left(\vec{v} \times \vec{B}\right)$

9. B
Sol. Motion of charged particle in a magnetic field $\vec{F} = q\left(\vec{v} \times \vec{B}\right)$

10. C
Sol. Motion of charged particle in a magnetic field $\vec{F} = q\left(\vec{v} \times \vec{B}\right)$

11. B
Sol. Charge Distribution

12. B
Sol. Charge Distribution

13. C
Sol. Charge Distribution

SECTION – B

14. 2
Sol. $E_{net} = \left(\frac{\sigma_1 - \sigma_2}{2\varepsilon_0}\right)$
Hence,
$W.D = \left(\frac{\sigma_1 - \sigma_2}{2\varepsilon_0}\right) q_0 \left(\frac{a}{\sqrt{2}}\right) = 2$

15. 2
Sol. Use motion of charge particle in magnetic field.

16. 4
Sol. $H = I^2R$

17. 2
Sol. Use Potential concept.

18. 7

Sol. Let,

E_1 and E_2 are electric field in x – 4 directions as shown:-

For q_1

$$\frac{V}{2}\cos 60^0 = V - \left(\frac{q_1E_1}{m_1}\right)t \quad ...(i)$$

$$\frac{V}{2}\sin 60^0 = 0 + \left(\frac{q_1E_2}{m_1}\right)t \quad .....(ii)$$

For q_2

$$0 = V - \left(\frac{q_2E_1}{m_2}\right)t \quad ....(iii)$$

From equation (i), (ii) and (iii).

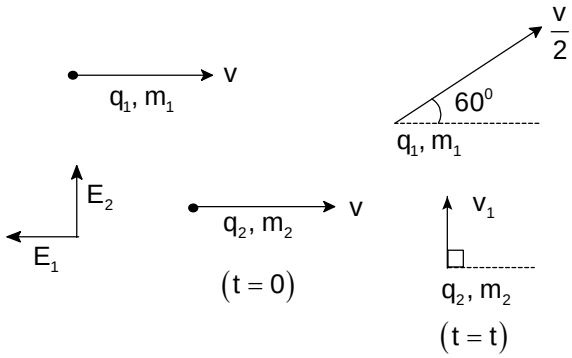
$$\frac{q}{m_2} = \frac{4}{3}\left(\frac{q_1}{m_1}\right) = \frac{4}{3}K$$

Hence,

$a = 4$

$b = 3$

$a + b = 7$



Chemistry

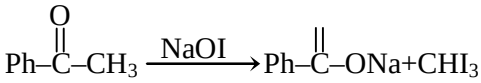
PART – II

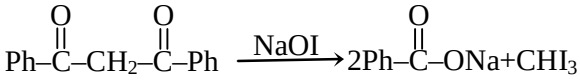
SECTION – A

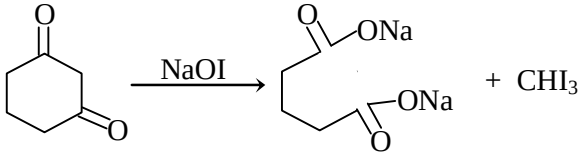
19. BCD

Sol. Concept of Williamson ether synthesis

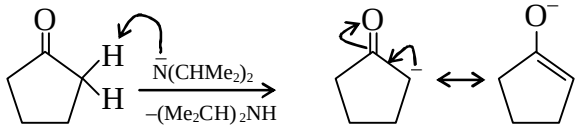
20. ABC

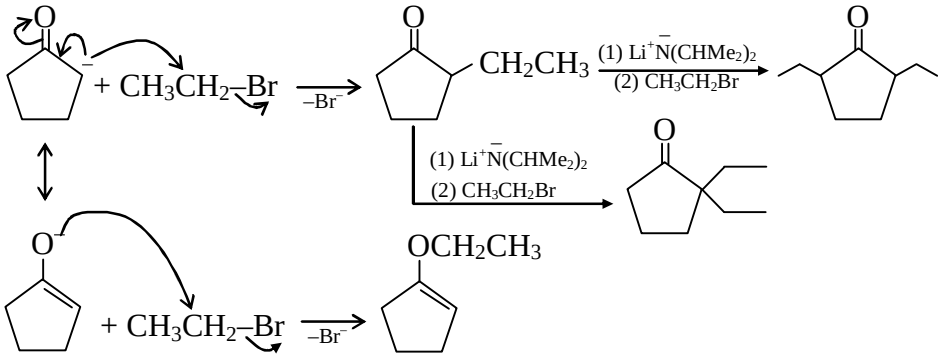
Sol. (A) 

(B) 

(C) 

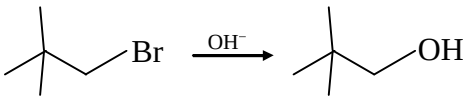
21. ABD

Sol. 

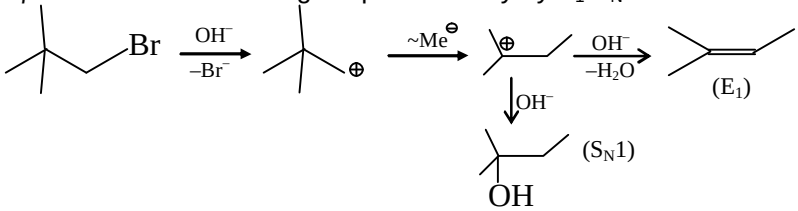


The base is not strong enough to cause any significant elimination.

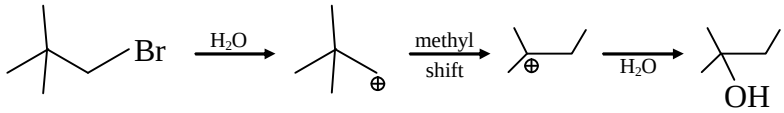
22. ABC

Sol. Reaction I: 

A primary (1°) substrate (alkyl halide) in presence of a strong nucleophile favours S_N2 reaction but substitution product formed is in small quantity due to substitution (crowding) at β-carbon. The reaction goes preferentially by E_1/S_N1 route.



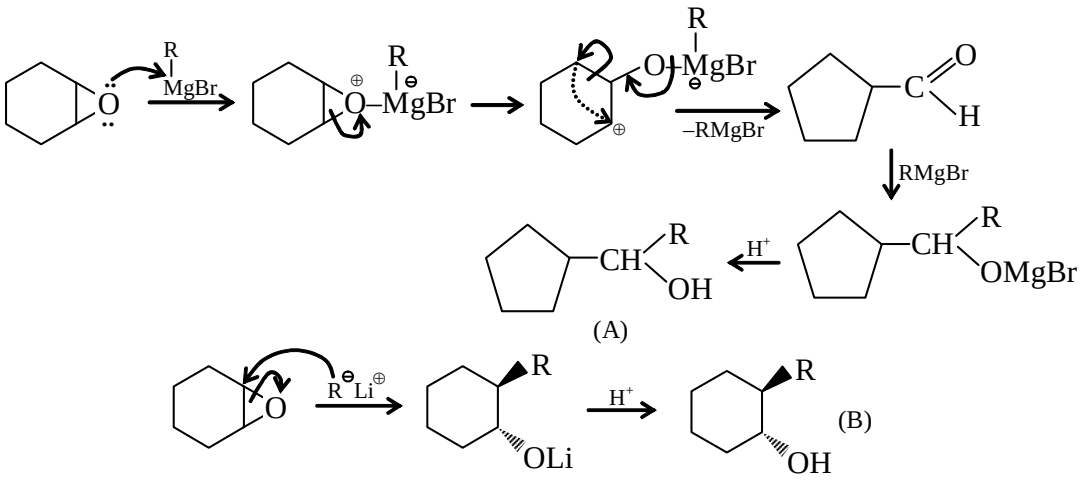
Reaction II:



Neopentyl bromide in presence of a weak nucleophile forms a carbocation that further gets rearranged to a more stable carbocation and thus shows S_N1 reaction.

23. AB

Sol.



24. BCD

Sol. $CH_2 = CH - CH_2 - CH = CH_2 \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} 2HCHO + H{-}\overset{O}{\overset{\|}{C}}{-}CH_2{-}\overset{O}{\overset{\|}{C}}{-}H$

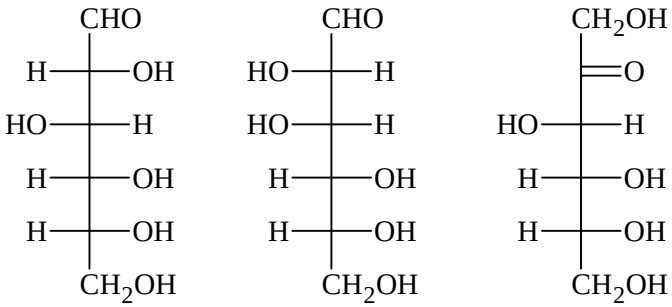
$CH_3{-}CH{=}\bigcirc \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} CH_3{-}CHO + \bigcirc{=}O$

$HC \equiv CH + H_2O \xrightarrow[H_2SO_4]{HgSO_4} CH_3{-}CHO$

$CH_3COCl + H_2 \xrightarrow[quinoline]{Pd\text{-}BaSO_4} CH_3{-}CHO$

25. ABC

Sol. The keto forms could be either of



26. A

Sol. Clemmensen reduction

27. A

Sol. Reaction of alcohol

28. B
Sol. Wolf – Kishner reduction

29. B
Sol. Concept of Nitration

30. C
Sol. Concept of sulphonation on aromatic compound

31. D
Sol. EAS

SECTION – B

32. 2
Sol. Moles of hydrocarbon (A) = Moles of Br_2 reacted

$\frac{0.7}{M} = \frac{2}{160}$, where M is the molecular mass of (A)

$\therefore M = \frac{0.7 \times 160}{2} = 56.$

The hydrocarbon (A) is C_4H_8 i.e., butene. Since it gives same product with HBr in presence as well as absence of peroxide, therefore, it is but–2–ene.

$$\underset{\text{(A)}}{CH_3–CH=CH–CH_3} + HBr \longrightarrow \underset{\text{(B)}}{CH_3–\underset{\displaystyle Br}{\underset{|}{CH}}–CH_2–CH_3}$$

Molecular weight of (B) = **137 g mol^{-1}**.
$137 \div 68.5 = 2$

33. 1
Sol. Let the mass of acetaldehyde is *x* g, then mass of ethanol is (0.535 – *x*)g. When acetaldehyde is heated with Fehling solution, it gives 1.2 g of a red precipitate.

$$CH_3CHO + 2CuO \rightarrow CH_3COO^- + \underset{\text{Red ppt.}}{Cu_2O}$$

$\because$ 143 g of Cu_2O is coming from = 44 g of CH_3CHO

$\therefore$ 1.2 g of Cu_2O is coming from $\frac{44}{143} \times 1.2$ = 0.37 g of CH_3CHO

% of $CH_3CHO = \frac{0.37}{0.535} \times 100$ = 69.16 %

$69.16 \times 100 \div 6916 = 1$

34. 6
Sol. $pH = \frac{PK_{a_1} + PK_{a_2}}{2}$

35. 4
Sol. Here, number of acidic hydrogens are 4.

36. 2
Sol. Hoffmann's degradation reaction.

Mathematics

PART – III

SECTION – A

37. ABC
Sol. Draw graph

38. D
Sol. $g'(x) = f'\left((\tan x - 1)^2 + 3\right)(2\tan x - 2)\sec^2 x$

$f'\left((\tan x - 1)^2 + 3\right) > 0 \quad \forall x \in (0, \frac{\pi}{4}) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$\therefore \tan x - 1 > 0 \Rightarrow x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

39. ABCD
Sol. Put x = 0 = y , f(0) = 1

$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = 2.f(x)$

40. C
Sol. $PN = x \Rightarrow ON = a - x$

$\Rightarrow ON \perp AB$ and AN = BN

$\Rightarrow$ ONP is $\perp$ r bisector of chord AB

$\therefore \cos\frac{\theta}{2} = \frac{|a-x|}{a}$

41. ABD
Sol. $\because f'(x) > 0, f(1) = 1 \Rightarrow f(x) > 1$

$\therefore f'(x) < \frac{1}{1+x^2} \forall x \in [1, \infty]$

Also, $f(x) = 1 + \int_1^x f'(t)dt < 1 + \int_1^x \frac{dt}{1+t^2}$

$= 1 + \tan^{-1}x - \frac{\pi}{4}$

42. ABCD
Sol. Find f'(x)

43. ACD
Sol. Put tanx = t

44. A
Sol. $f(-x) = f(x) \Rightarrow f(x)$ is even

$f(-x) = -f(x) \Rightarrow f(x)$ is odd

Use the concept of derivate to decide f(x) is one-one or many one

45. C
Sol. $f(-x) = f(x) \Rightarrow f(x)$ is even

$f(-x) = -f(x) \Rightarrow f(x)$ is odd

Use the concept of derivate to decide f(x) is one-one or many one

46. D

Sol. $f(-x) = f(x) \Rightarrow f(x)$ is even

$f(-x) = -f(x) \Rightarrow f(x)$ is odd

Use the concept of derivate to decide f(x) is one-one or many one

47. B

Sol. Calculate LHL and RHL individually at x = 0

48. B

Sol. Calculate LHL and RHL individually at x = 0

49. A

Sol. Calculate LHL and RHL individually at x = 0

SECTION – B

50. 3

Sol. Let $\lim_{n\to\infty} f(n) = x \Rightarrow \lim_{n\to\infty} f(n+1) = x$

$\therefore x = \frac{1}{2}\left(x + \frac{9}{x}\right) \Rightarrow x = 3$

51. 4

Sol. $F(x) + F\left(x + \frac{1}{2}\right) = 3 \longrightarrow$(i)

Replace

x. by $x + \frac{1}{2}$

$F\left(x + \frac{1}{2}\right) + F(x+1) = 3 \longrightarrow$(ii)

(i) – (ii) $\Rightarrow F(x) = F(x+1)$

$\therefore$ F(x) is periodic with period 1

So,

$I = \int_0^{1500} F(x)dx$

$= 1500\left[\int_0^{1/2} F(x)dx + \int_{1/2}^{1} F(x)dx\right]$

$= 1500\left[\int_0^{1/2} F(x)dx + \int_0^{1/2} F\left(x + \frac{1}{2}\right)dx\right]$

$= 2250$

52. 3

Sol. $(gof)\left(g\left(\frac{2}{3}\right)\right) = (gof)(-1)$

$= g\left(f\left(-1\right)\right)$

$= g\left(2\right) = 2\times 2-1 = 3$

53. 6

Sol. $f'\left(x\right) = 6x^2 + 2ax + b$

$a^2 > 6b \Rightarrow a = 3, b = 1$

54. 1

Sol. As $x \to 0^-$, $\left[x\right] = -1$

$\therefore \lim\limits_{x\to 0^-} f\left(x\right) = \lim\limits_{x\to 0^-} \dfrac{1+\sin\left(-1\right)}{-1}$

$= -1+\sin 1$